

Azure Capacity Reservation Management Engine (ACRME)

Executive Design Document — Prepared for Leadership Review

Booking, sharing, and protecting cloud capacity across customers and regions — with disaster recovery built in.

The Position in One Slide

What ACRME Is

A central control system that books, shares, and protects cloud server capacity across many customer accounts and locations — a reservation desk for cloud computing power.

What It Does

- Maintains shared pools of reserved capacity
- Chooses the best location for each customer
- Protects a dedicated disaster-recovery floor
- Requires human approval for high-risk actions
- Records every decision for audit

Design

Complete & independently reviewed

Fact-check

7 gaps identified and corrected

Validation

Planned — not yet executed end-to-end

Production readiness

≈ 4 / 10 — strong design, unproven execution

The Business Problem

Cloud capacity is only guaranteed when it is reserved in advance. Reserved capacity is billed whether or not it is used — so over-reserving wastes budget and under-reserving creates service risk. Managing that trade-off manually across dozens of customers and regions does not scale.

What goes wrong without a system

Service disruptions when capacity is unavailable at the moment of need

Failed or severely delayed disaster recoveries — recovery capacity was never secured

Wasted cloud spend from over-reservation nobody is systematically reviewing

Inconsistent practices across teams, making audit and compliance difficult

No single view of what is reserved, used, or protected for recovery

The Three Types of Capacity

ACRME keeps three classes of reserved capacity separate in every location — so everyday demand can never quietly consume what disaster recovery depends on.

Production	Non-production	Disaster recovery
Purpose Everyday customer workloads that must always run How it is treated Always reserved and paid for. Highest day-to-day priority.	Purpose Development, testing, and staging How it is treated Reserved but flexible. Must not crowd out the DR share.	Purpose Protected floor in a different location How it is treated Most protected. Engine-enforced floor. Breaching actions blocked.

Hard rule: a customer's production, non-production, and DR workloads must not all sit in one geography. Production and DR must be in different locations.

The Automation Engine — Safety Before Automation

The engine continuously:

- Monitors reservation levels, usage, and headroom
- Decides when capacity should be increased
- Routes customers to the best location by rules & policy
- Prevents any action from destroying the DR floor
- Requires human approval for the riskiest operations
- Keeps a complete audit trail of every change

Operations are tiered by risk

Tier 1 — Additive

Add DR capacity from pre-staged headroom

Allowed in a declared disaster, with fresh checks

Tier 2 — Reallocation

Reduce non-prod to expand DR

Disabled until validation proves behaviour

Tier 3 — Disassociation

Change how servers link to reservations

Blocked — needs permissions model + board sign-off

How Placement Decisions Are Made

Managed regions are split into two classes. Hard rules filter candidates first; the engine then scores the survivors on five weighted components.

Standard Capacity Regions

North America: West US 3, Central US, Canada Central

Europe: Sweden Central, Belgium Central

Middle East: Saudi Arabia, UAE North

Asia Pacific: Japan East, Southeast Asia, Australia East

Restricted Capacity Regions

Excluded from automated placement — eligible only via an explicit, approved exception (production only).

- East US 2 (North America)
- North Europe, West Europe (Europe)
- East Asia, Australia Southeast (Asia Pacific)

Scoring components

Available capacity · Quota headroom · Distribution fairness · DR coverage health · Zone diversity

Weights are versioned and recorded with every decision. During the pilot, placement is recommendation-only — a human commits.

Middle East — Cross-Geo DR Extension

Middle East has only two Standard Capacity Regions, but a full deployment needs three (production, non-production, disaster recovery). The engine extends DR to Belgium Central in Europe — the only currently approved cross-geography DR path.

Production	Higher-scoring of Saudi Arabia or UAE North
Non-production	The other in-geography region (deterministic)
Disaster recovery	Belgium Central (Europe) — cross-geo

Belgium Central must still pass capacity and quota checks; if it fails, the deployment is blocked with an alert — the engine never silently substitutes another region.

How Disaster Recovery Works

Declaration & failover

- Authorised operator requests declaration
- Dual approval + state validation required
- Engine enters disaster-event mode
- Capacity, sharing, zones, quota re-validated
- Approved failover deployments begin, observed per workload
- Incident-hold mode if state conflicts appear

The protected capacity floor

A calculated minimum that must stay available for recovery (design placeholder 30-40%). Enforced by ACRME — not a native cloud sub-reservation.

An independent detector recalculates the floor and blocks non-production expansion if the two calculations disagree (fail-closed).

Failback is gated

Controlled return to primary only after explicit approval, readiness validation, wave-based restore, and health checks. Starting too early is treated as a critical risk.

What Is Working Well

Comprehensive, reviewed architecture

Broad, coherent, independently reviewed; strong requirements coverage

Complete risk register

44 risks logged with likelihood, impact, mitigation, owner, residual

Phased approach with human control

Dangerous operations blocked or approval-gated in Phase 1

Strong audit & observability design

Append-only audit records; dashboards and critical guard alerts

Fact-check corrections applied

7 corrections; high-severity gaps written back into the design

Manual survivability

Operators can still recover if the engine is unavailable

What Is Still Being Validated

None of these should be described as solved until the stated validation is done.

BLOCKER

Permissions model

Number-one production blocker — security review & approval required

BLOCKER

End-to-end DR untested

Failover/failback not yet run live — go-live blocker

VALIDATING

Shared reservation is Preview

Not covered by GA commitments; can change or be withdrawn

VALIDATING

100-customer pool ceiling

Hard platform limit; pools must be sharded beyond it

VALIDATING

Zone alignment across accounts

Mandatory onboarding gate; needs execution proof per pair

VALIDATING

VMSS DR via shared reservation

Preview limitation; blocked for automated Phase 1 use

The Risks — Plain English

44 logged risks across four dimensions. Highest-impact items and what the programme is actually doing:

Technology Risks

- Preview feature changed/withdrawn → feature flags, governance gate
- VMSS can't reprovision in outage → block Phase 1, require GA path
- Stale inventory drives a bad decision → confirm with direct reads

Operational Risks

- Permissions model over/under-grants → scoped MI, Tier 3 blocked
- DR fails when first exercised → full exercise is a production gate
- DR floor bypassed by bug/stale data → independent fail-closed detector

Commercial Risks

- Discounts assumed that don't apply → per-customer scope validation
- Over-reservation persists in Phase 1 → accepted cost of safety
- DR coverage ratio mis-set → 30-40% is placeholder; needs policy

Customer Risks

- Zone mismatch makes reservation useless → mandatory zone map
- 100-account ceiling breached → shard pools, monitor continuously
- Forced pool removal strands restarts → default-deny + runbook

Three Decisions Leadership Must Make

1

Accept the Preview feature dependency

Shared reservations are still Microsoft Preview. Recommendation: accept for a bounded pilot after gates; require a separate production decision later.

2

Approve the permissions model approach

The credential model is the #1 production blocker. Recommendation: customer-consented managed identity with narrow, resource-group-scoped roles — never subscription-wide admin. Tier 3 stays blocked until security approves.

3

Set the disaster recovery floor policy

How much capacity is held for DR is a business decision. 30–40% is a placeholder. Recommendation: customer-specific recovery analysis for critical workloads.

Until each decision is made explicitly, the programme is not cleared for the corresponding scope.

Recommended Go-Live Sequence

Three stages, each with entry gates. Skipping gates exports risk to customers.

Phase 1 Pilot (Constrained)	Phase 2 Controlled Automation	Production Full Automation (After Gates)
• Few customers; manual approval for capacity changes • Placement recommendation-only; DR automation off • Destructive & VMSS emergency actions rejected • Gate: Preview risk accepted, zones validated, rollback proven	• Automate lower-risk operations on Phase 1 evidence • DR automation remains gated; reallocation only if proven • Tier 3 blocked until permissions + state machine close • Gate: concurrency proven, mode transitions fault-tested	• Only after all gates pass and blockers close • DR tested end-to-end with recovery objectives met • Cost/discount validated per customer vs. billing • Destructive tiers need separate board authorisation

Council position: approve a two-stage programme — constrained pilot now, controlled production later. Unrestricted autonomous DR is not approved.

Immediate Actions for Leadership

- 1 Record a formal Preview-feature decision for the pilot, with a named owner of residual risk
- 2 Commission the security review of the customer permissions model and set a decision date
- 3 Set interim DR floor and emergency headroom policy values for the pilot cohort
- 4 Confirm the pilot customer list and written acceptance of known limitations
- 5 Schedule the end-to-end DR exercise as a gated milestone on the production path

The honest position: design complete, validation in progress, conditional pilot supportable — unrestricted autonomous disaster-recovery automation not approved.